

Assignment - 4

1) construct nfa with E loops equivalent to the R.E given below.

1) $(a^* + b^*)^*$

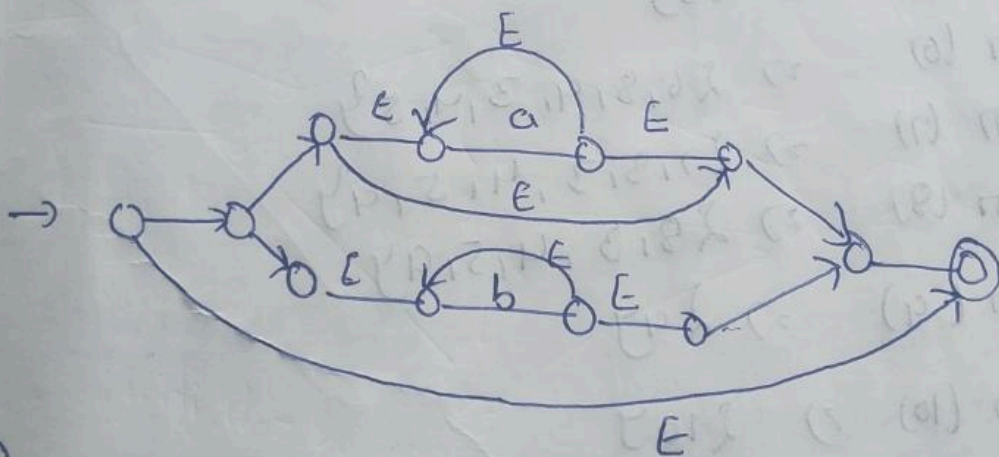
2) $(01+10)^*$

3) ab^*a

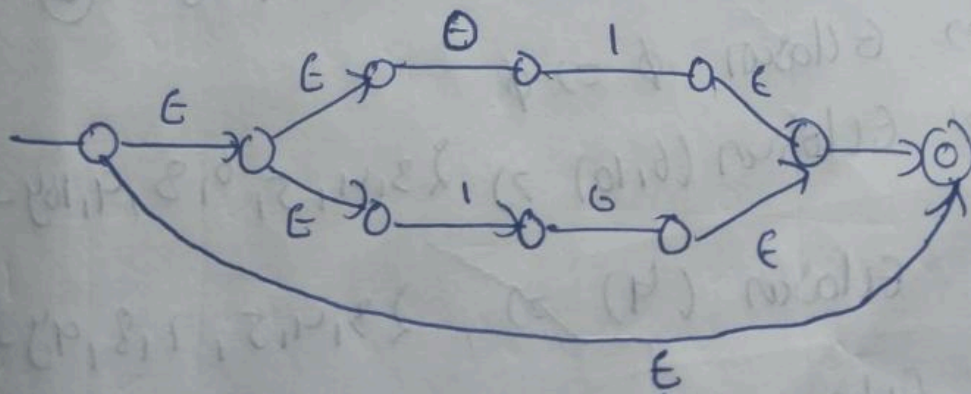
2) convert RE $a(a+b)^*a$ into E-nfa and find the minimal state of DFA.

Answers:

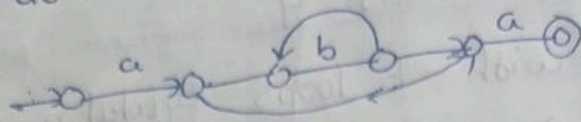
1) RE $= (a^* + b^*)^*$



2) $(01+10)^*$

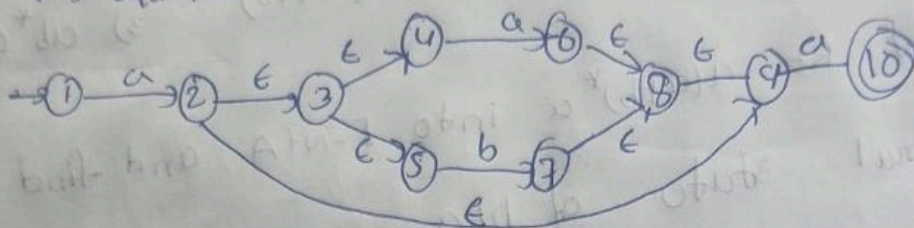


① ab^*a



②

Ref $\Rightarrow a(a+b)^*a$ into DFA and minimize.



ϵ closure (1) $\Rightarrow \{1\} \rightarrow A$

ϵ closure (2) $\Rightarrow \{2, 3, 4, 5, 9\}$

ϵ closure (3) $\Rightarrow \{3, 4, 5\}$

ϵ closure (4) $\Rightarrow \{4\}$

ϵ closure (5) $\Rightarrow \{5\}$

ϵ closure (6) $\Rightarrow \{6, 8, 9, 3, 4, 5\}$

ϵ closure (7) $\Rightarrow \{7, 8, 3, 4, 5, 9\}$

ϵ closure (8) $\Rightarrow \{8, 3, 4, 5, 9\}$

ϵ closure (9) $\Rightarrow \{9\}$

ϵ closure (10) $\Rightarrow \{10\}$

$\delta(A, a) \Rightarrow \epsilon$ closure 2 $\Rightarrow \{2, 3, 4, 5, 9\} \rightarrow B$

$\delta(A, b) \Rightarrow \epsilon$ closure $\emptyset \Rightarrow \emptyset$

$\delta(B, a) \Rightarrow \epsilon$ closure (6, 10) $\Rightarrow \{3, 4, 5, 6, 8, 9, 10\} \rightarrow C$

$\delta(B, b) \Rightarrow \epsilon$ closure (4) $\Rightarrow \{3, 4, 5, 7, 8, 9\} \rightarrow D$

$\delta(C, a) \Rightarrow \epsilon$ closure (4, 9) $\Rightarrow \{4, 9\} \rightarrow E$

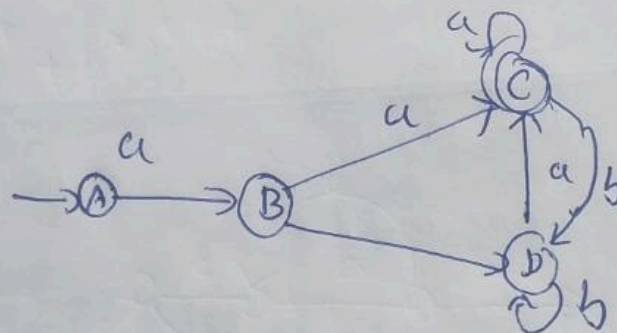
$\delta(C, b) \Rightarrow \epsilon$ closure (7) $\Rightarrow \{7\} \rightarrow F$

$\delta(D, a) \Rightarrow E$ (odd) $(b, 10) \Rightarrow (c)$
 $\delta(D, b) \Rightarrow E$ (odd) $(1) \Rightarrow (b)$
 $\delta(E, a) \Rightarrow E$ (odd) $(b, 10) \Rightarrow (c)$
 $\delta(E, b) \Rightarrow E$ (odd) $(\phi) \Rightarrow \phi$

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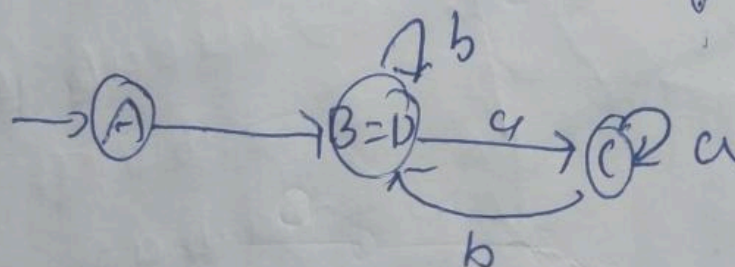
	a	b
A	B	ϕ
B	C	D
C	E	D
D	C	D
E	C	D

DFA



$B = D$

minimized DFA



8/2